

Sahil Khan

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EDUCATION

Drexel University, School of Biomedical Engineering | Philadelphia, PA

May 2028

B.S. in Biomedical Engineering and MS in Biomechanics

Relevant Coursework: Mechanics, Dynamics, Statics, Advanced Biocomputational Languages, Thermodynamics and Kinetics, Circuits, Signals and Systems, Biocomputational Modeling and Simulation

SKILLS

Programs: NX, Solidworks, Onshape, AutoCad, Autodesk Fusion 360, Creo, MATLAB, Simulink, Simscape, Python, Java, C++, Rust, JavaScript, SAP, Cornerstone, Altium, Multisim, Figma, Arduino IDE **Machine Learning:** TensorFlow, NumPy, Pandas, Matplotlib **Tools & Technologies:** Git, GitLab, Linux/Unix, VS Code, Jupyter Notebook

EXPERIENCE

US Naval Research Laboratory

Washington DC

Spacecraft Mechanisms Design Engineer Co-op

September 2025 - March 2026

- Debugged a flight controller algorithm using MATLAB/Simulink, including model-based simulation, PID, parameter tuning, and stability/ performance verification
- Designed actuatable robotic structures using NX; integrated motors/actuators and fabricated designed components via additive manufacturing
- Designed and iterated test-fixture and experimental setup assembly models in Siemens NX for fabrication
- Developed a Python-based application that interfaces with the NX API to automate the generation and downloading of STEP-format CAD files, directly supporting rapid prototyping efforts for hypersonic vehicles

Estee Lauder Companies

Bristol PA

Manufacturing Engineer (Commercialization and Validation)

September 2024 - March 2025

- Improved product quality and reduced manufacturing costs by 15%, conducting process validation on production lines and optimizing workflows and equipment performance
- Created detailed Process Flow Diagrams and Transformative Flow Sheets to streamline and document processes
- Ensured compliance across 250+ products by performing validation testing on formulations, packaging, and manufacturing processes, adhering to FDA, EMEA, and internal quality standards
- Prepared and submitted regulatory documentation via SAP and Cornerstone, keeping up with KPIs and ensuring timely approvals for product launches worth \$1M+ across global markets

RESEARCH / PROJECTS

Zheng Research Lab | Independent Researcher

June 2024 - Present

- Designed Tetra-chiral structures to improve shock absorption and lightweight strength in prosthetic legs, utilizing SolidWorks for CAD modeling, 3D printing, ASTM-standard testing, and Finite Element Analysis (FEA)
- Analyzed mechanical properties and dimensional variations, identifying 2x2x10 tessellations as the optimal configuration for load distribution and force-to-displacement efficiency
- Presented findings at Drexel University's College of Medicine, highlighting advancements in prosthetic technology, proposing applications to enhance durability, user comfort, and structural performance

Modular Prosthetic with EMG Intent Prediction (J&J Engineering Showcase)

August 2025- April 2026

- Developed an EMG signal-processing pipeline to apply Butterworth band-pass filtering, rectification, moving-window RMS smoothing, ADC conversion, and MVC normalization for muscle activation detection
- Designed a modular 3D-printed prosthetic hand assembly in Solidworks with pivot joints, linear actuation, alloy steel bearings, and swappable end-effector components
- Integrated embedded control architecture using Raspberry Pi Zero 2 W, ATmega32U4 motor control, lithium-ion battery power, BMS protection, and regulated DC-DC power delivery

LEADERSHIP & INVOLVEMENT

Drexel Institute of Electrical and Electronics Engineers (IEEE)

Philadelphia PA

President

Sep 2024 - Present

- Lead and manage a team of 5 executive board members to organize professional, technical, and outreach events for over 700+ registered members, hosting weekly meetings and monthly general bodies
- Oversaw budgeting and secured sponsorships, increasing funding by 50% through external partnerships
- Hosted Drexel's annual 24 hour hackathon along with MLH, hosting over 200 participants and recruiters from companies like Amazon, EY, DoorDash, NASDAQ, and Coinbase.